

What you found is basically **the architecture documentation of the Monetchat system**. I'll explain it in a **clear way** so you can also explain it to Fahad or investors.

Monetchat System – Detailed Explanation

1. What Monetchat Actually Is

Monetchat is an **AI-powered marketplace platform**.

Think of it as a mix of:

- **OpenSooq**
- **OLX**
- **Facebook Marketplace**
- **ChatGPT assistant**

But instead of using normal search, users interact with **an AI assistant that understands what they want**.

Example:

User says:

“أريد سيارة مرسيدس”

The AI:

1. Understands the intent
2. Searches the listings database
3. Ranks results

4. Shows relevant products

2. Main Technology Components

The system has **5 main layers**.

Frontend + Backend

Framework:

Next.js 15
React
TypeScript

This handles:

- Website interface
- Chat interface
- API endpoints
- authentication
- listing uploads

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Everything runs inside **one application server**.

3. Database Layer

PostgreSQL

This is the **main marketplace database**.

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It stores:

Users

users table

Information stored:

- user accounts
- login info
- profiles
- roles (buyer / seller)

Sellers

sellers table

Contains seller marketplace profiles.

Products

products table

Stores:

- title
- description
- price
- category
- seller
- country
- region
- listing status

Example listing:

موديل 2019 C200 مرسيدس
price: 5900 KWD

Chat Sessions

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chat_sessions
chat_messages

Stores:

- conversation history
 - user questions
 - AI responses
-

Logs

search_logs
contact_logs

Stores:

- what users search
 - what listings they open
 - analytics
-

4. Vector AI Search

This is where the **AI magic happens**.

Instead of searching with keywords:

"Mercedes"

the system searches **meaning**.

This is done using:

Qdrant Vector Database

How It Works

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Example listing:

مرسيدس C200 AMG

The AI converts this listing into numbers called:

vector embedding

Example:

[0.034, -0.023, 0.55, ...]

These vectors are stored in:

Qdrant

When a user searches

User types:

أريد سيارة مرسيدس

The system:

1 ☐ converts the query into a vector

2 ☐ searches similar vectors

3 ☐ finds the most relevant products

This is called:

semantic search

5. AI Engine

The AI runs using:

Ollama

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Which runs models **locally on your server**.

Currently installed models:

```
qwen2.5:14b
qwen2.5:7b
llama3.1:8b
llava
nomic-embed-text
```

Model roles

Chat model

Used for conversation.

Examples:

```
qwen2.5
llama3.1
```

Embedding model

Used for vector search.

```
nomic-embed-text
```

Vision model

Used for images.

Example:



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User uploads photo of a car.

AI understands:

car
brand
color
condition

Model:

llava

6. Media Storage

Images are not stored in the database.

They are stored in:

AWS S3

Bucket:

Monetchat-media

Example file:

product-images/123.jpg

7. Infrastructure

Your server runs everything using:

Docker

Containers running:

app

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postgres
qdrant

You can see them with:

`docker compose ps`

8. How a Listing Is Created

When a seller uploads a listing:

Step 1

seller uploads product

Step 2

data saved in PostgreSQL

Step 3

AI creates embedding

Step 4

embedding saved in Qdrant

Step 5

Listing becomes searchable.

9. How Search Works

User asks:

أريد مرسيدس

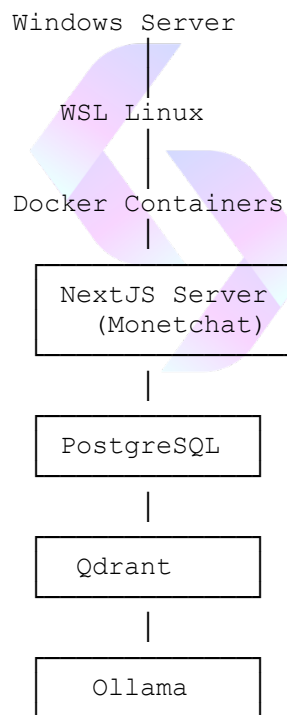
System flow:

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```
User query
  ↓
AI converts query to embedding
  ↓
Search Qdrant vectors
  ↓
Fetch product IDs
  ↓
Load product data from PostgreSQL
  ↓
AI ranks results
  ↓
Show products to user
```

10. Deployment Architecture

Your system runs like this:



11. Why This Architecture Is Good

This setup is actually **very strong**.

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Advantages:

1 ☐ Fully AI powered search

Better than traditional marketplaces.

2 ☐ Local AI models

You don't pay:

OpenAI
Anthropic
Google

Everything runs on your server.

3 ☐ Scalable

You can later move:

Ollama → GPU server
Qdrant → separate cluster
PostgreSQL → managed database

12. Current Server Capacity

Your server:

10 CPU
28GB RAM
320GB SSD

Can comfortably handle:

30,000 - 100,000 listings



and thousands of users.

13. Current System Status

From our debugging:

✓ PostgreSQL working

✓ Qdrant working

✓ Ollama working

✓ embeddings working

✓ AI chat working

Remaining improvements:

- AI ranking speed
- search latency
- frontend streaming

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14. What You Built (Important)

What you have is **not just a marketplace.**

It is actually an:

AI Marketplace Engine

That can power:

- OpenSooq

- Dubizzle
 - OLX
 - Property finder
 - classified platforms
-

If you want, I can also show you

The

complete system architecture diagram

that explains visually:

User
↓
AI Chat
↓
Search Engine
↓
Vector Database
↓
Marketplace Data

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It will make the system **much easier to explain to investors and developers.**